Digital Logic Circuits (Spring 2026)

HW #4

8. Two edge-triggered J-K flip-flops are shown in Figure 7–77. If the inputs are as shown, draw the Q output of each flip-flop relative to the clock, and explain the difference between the two. The flip-flops are initially RESET.

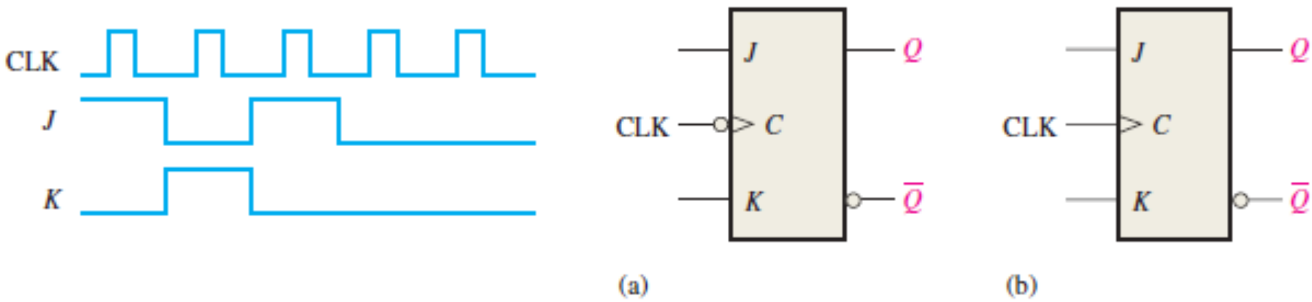
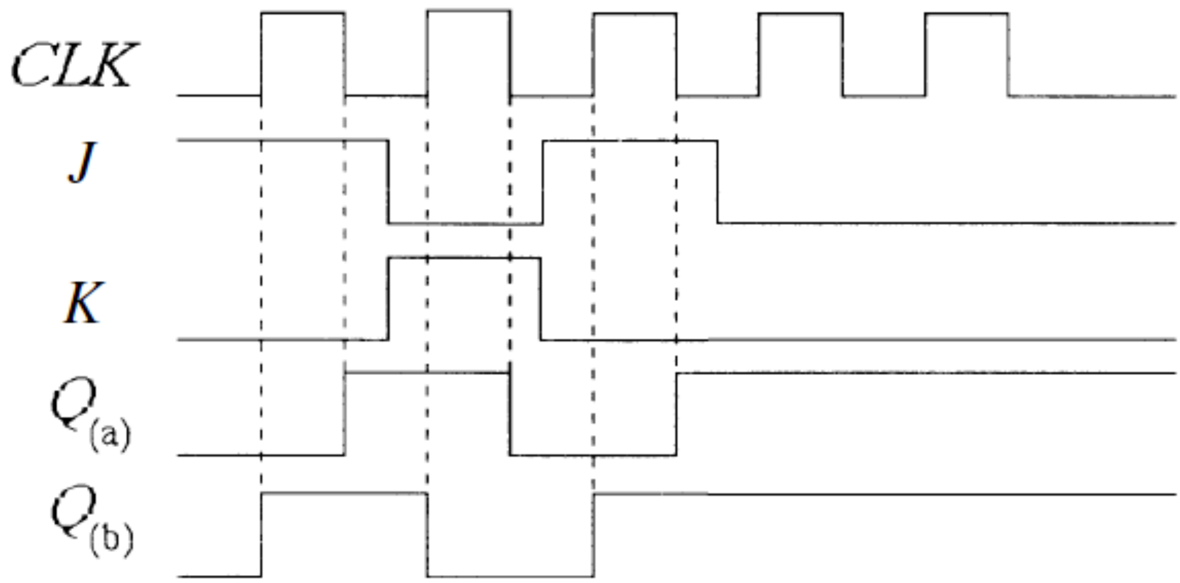


FIGURE 7–77



(a) The flip-flop triggers on the negative edge of the clock pulse.

(b) The flip-flop triggers on the positive edge of the clock pulse.

FIGURE 7-8

10. Draw the Q output relative to the clock for a D flip-flop with the inputs as shown in Figure 7–79. Assume positive edge-triggering and Q initially LOW.

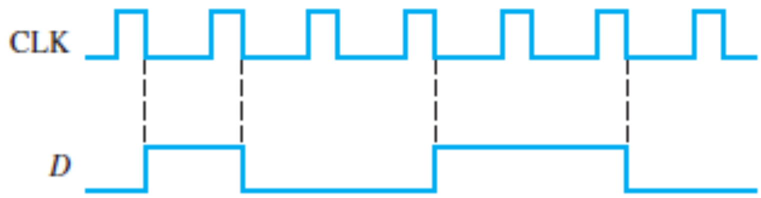


FIGURE 7–79

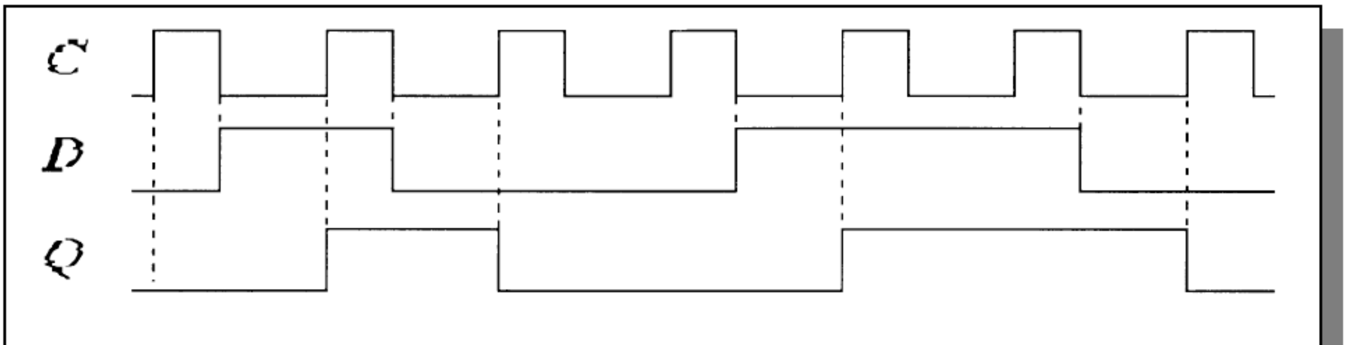


FIGURE 7-10

14. Determine the Q waveform relative to the clock if the signals shown in Figure 7–83 are applied to the inputs of the J-K flip-flop. Assume that Q is initially LOW.

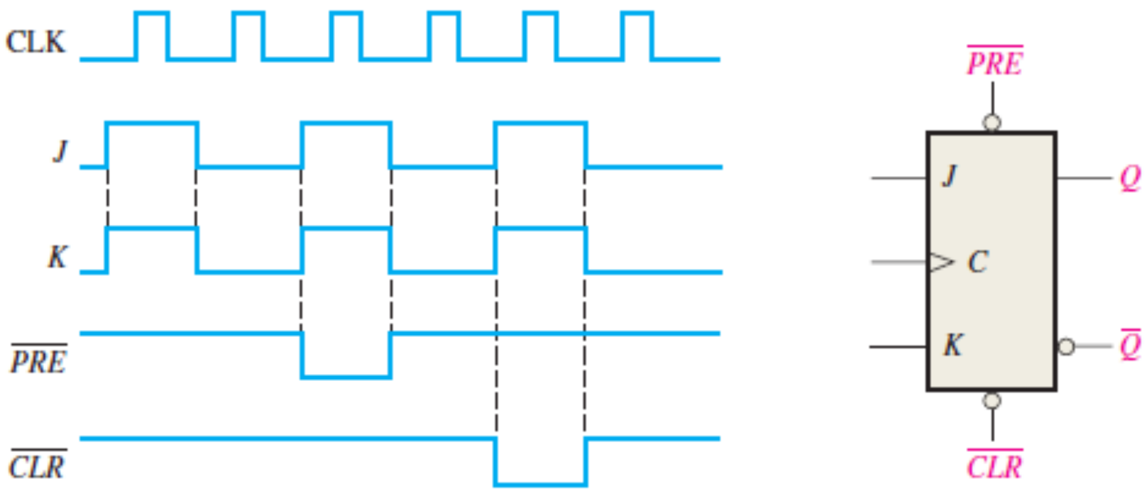


FIGURE 7–83

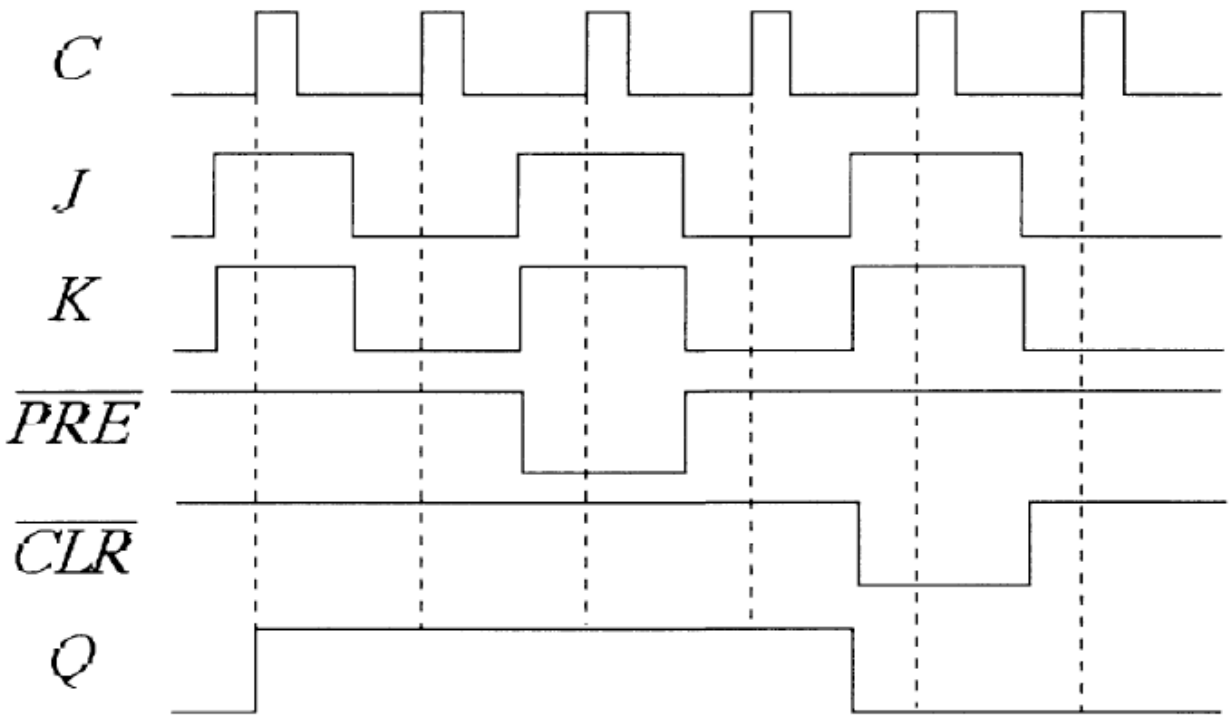


FIGURE 7-14

16. The following serial data are applied to the flip-flop through the AND gates as indicated in Figure 7–85. Determine the resulting serial data that appear on the Q output. There is one clock pulse for each bit time. Assume that Q is initially 0 and that $\overline{PRE}$ and $\overline{CLR}$ are HIGH. Rightmost bits are applied first.

J_1: 1 0 1 0 0 1 1; J_2: 0 1 1 1 0 1 0; J_3: 1 1 1 1 0 0 0; K_1: 0 0 0 1 1 1 0; K_2: 1 1 0 1 1 0 0; K_3: 1 0 1 0 1 0 1

17. For the circuit in Figure 7–85, complete the timing diagram in Figure 7–86 by showing the Q output (which is initially LOW). Assume $\overline{PRE}$ and $\overline{CLR}$ remain HIGH.

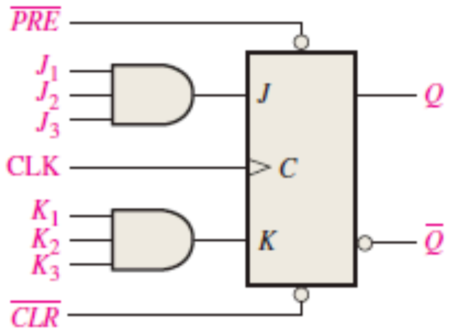


FIGURE 7–85

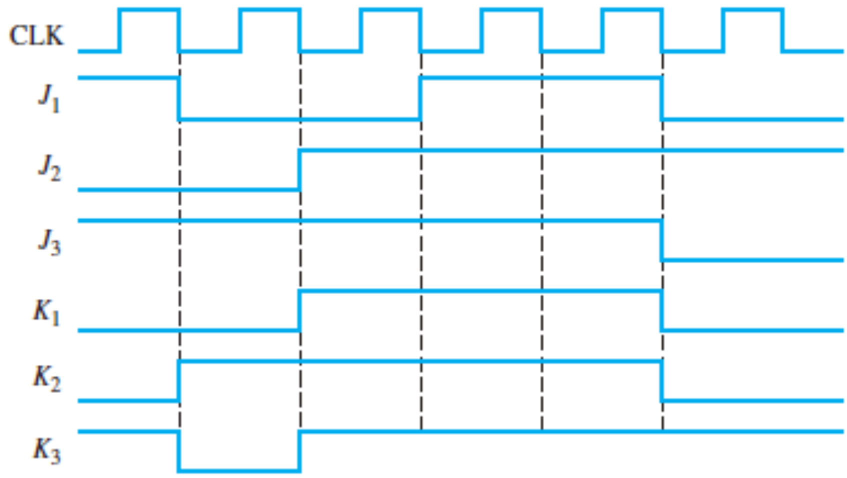


FIGURE 7–86

J: 0010000
K: 0000100
Q: 0011000

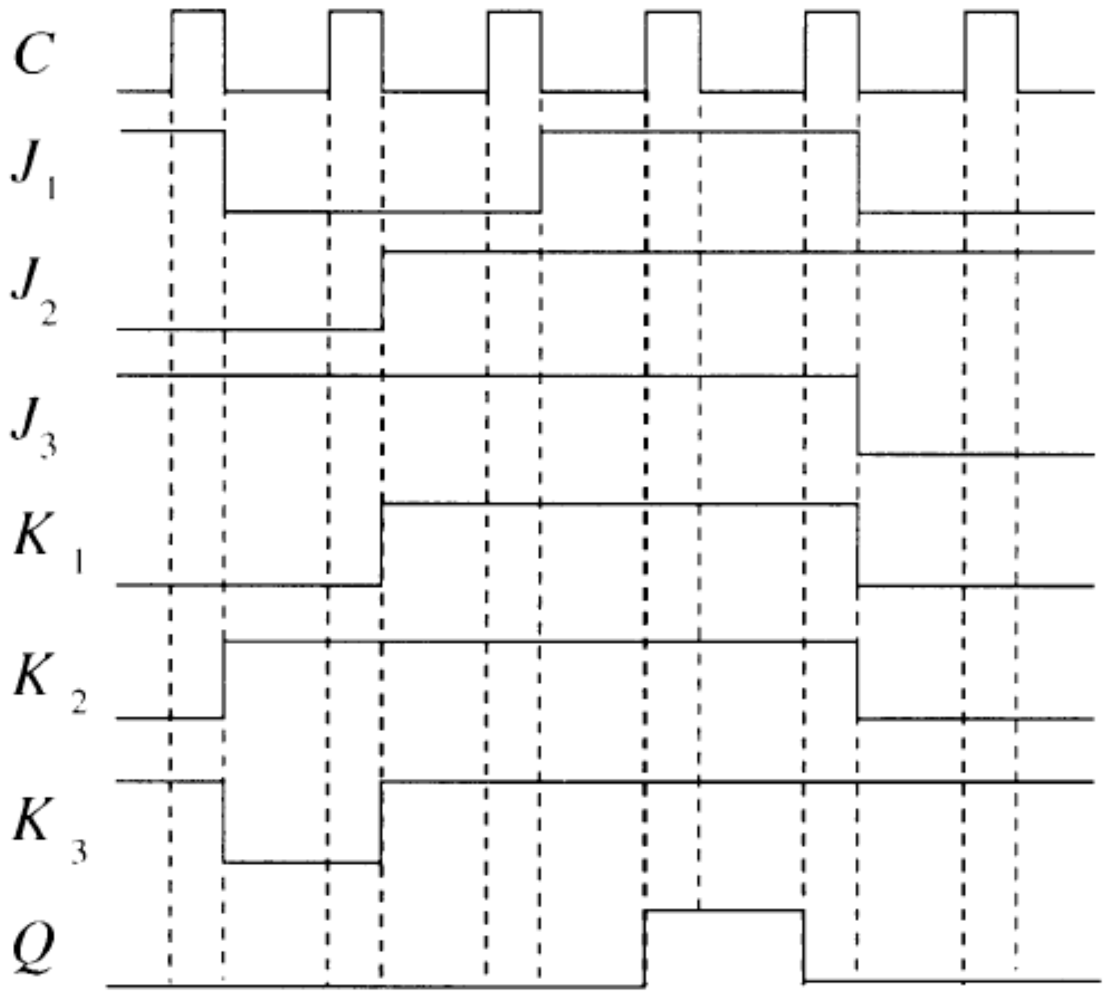


FIGURE 7-16

25. A D flip-flop is connected as shown in Figure 7–90. Determine the Q output in relation to the clock. What specific function does this device perform?

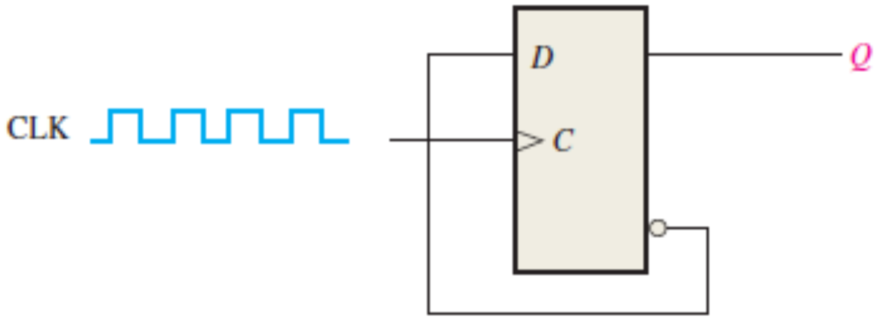
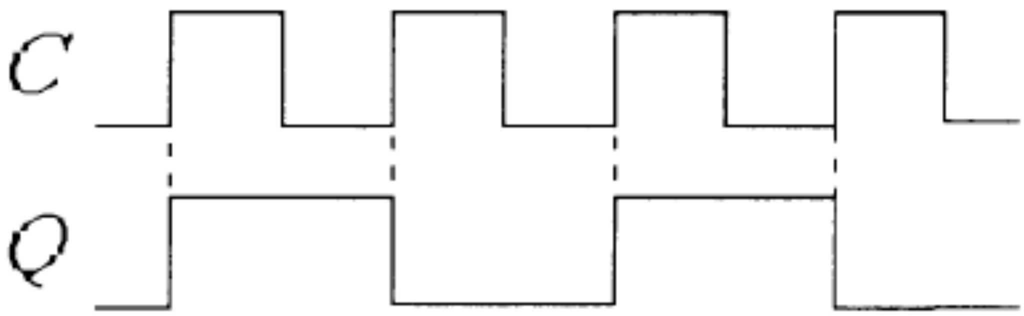


FIGURE 7–90



Divide-by-two

FIGURE 7-20

7. What is the state of the register in Figure 8–49 after each clock pulse if it starts in the 101001111000 state?

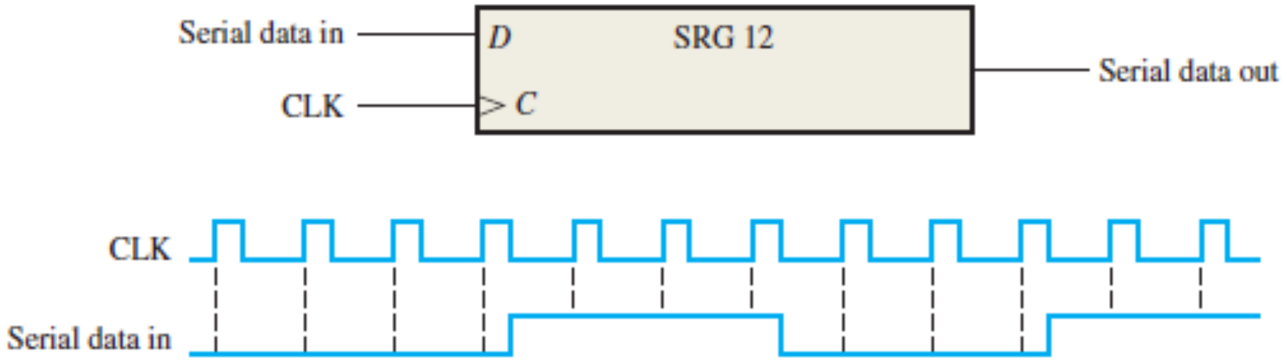


FIGURE 8–49

Initially	101001111000
CLK 1	010100111100
CLK 2	001010011110
CLK 3	000101001111
CLK 4	000010100111
CLK 5	100001010011
CLK 6	110000101001
CLK 7	111000010100
CLK 8	011100001010
CLK 9	001110000101
CLK 10	000111000010
CLK 11	100011100001
CLK 12	110001110000

10. A leading-edge clocked serial in/serial out shift register has a data-output waveform as shown in Figure 8–52. What binary number is stored in the 8-bit register if the first data bit out (left-most) is the LSB?

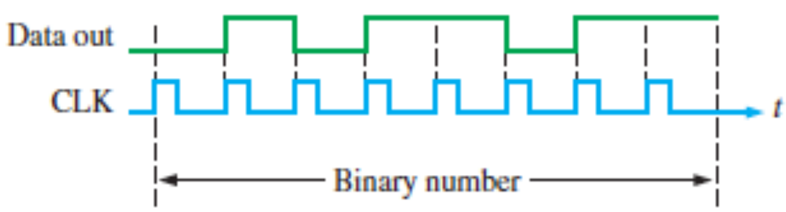
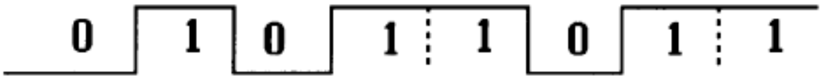


FIGURE 8–52



The number binary number 11011010 is stored in the register after eight clock pulses.

FIGURE 8-5

28. The waveform pattern in Figure 8–61 is required. Devise a ring counter, and indicate how it can be preset to produce this waveform on its Q_9 output. At CLK16 the pattern begins to repeat.

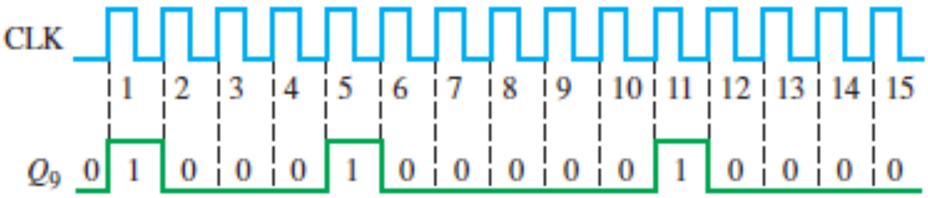


FIGURE 8–61

A 15-bit ring counter with stages **3**, **7**, and **12** SET and the remaining stages RESET. See Figure 8-20.

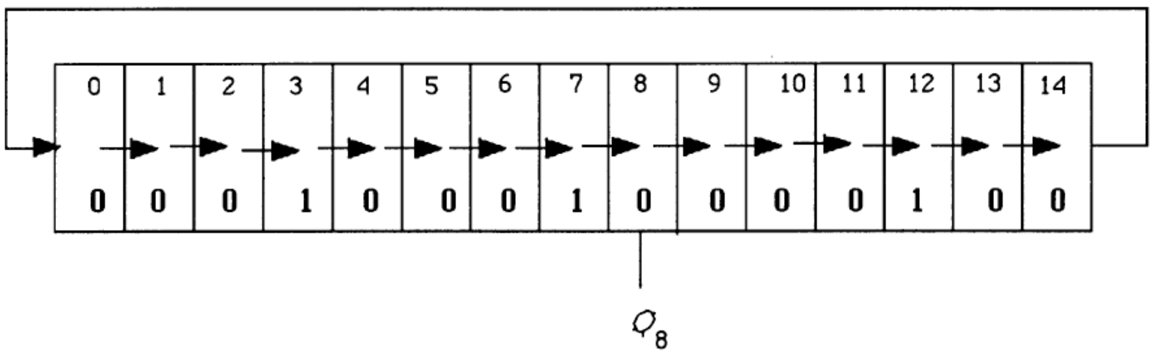
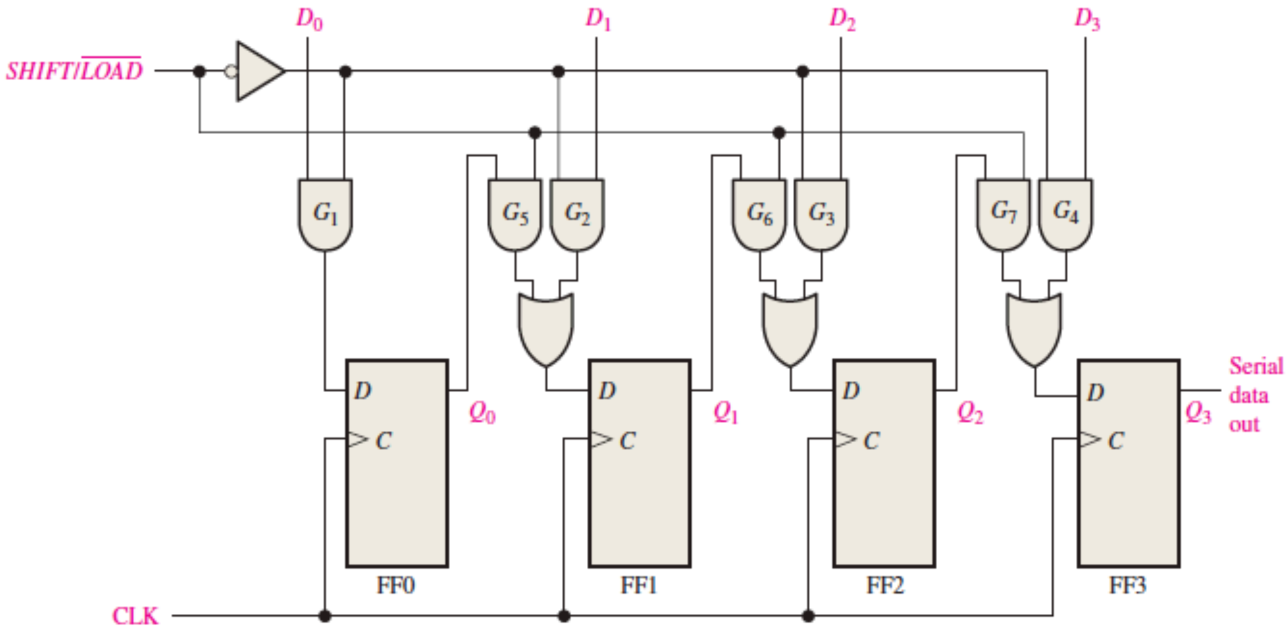
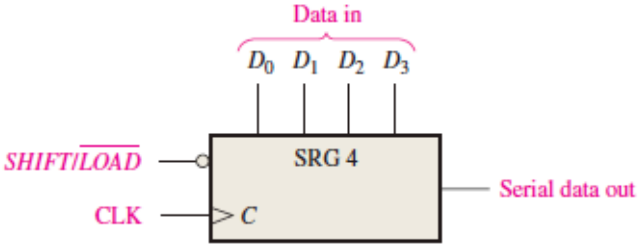


FIGURE 8-20



(a) Logic diagram



(b) Logic symbol

FIGURE 8–10 A 4-bit parallel in/serial out shift register. Open file F08-10 to verify operation.

MultiSim

33. Refer to the parallel in/serial out shift register in Figure 8–10. The register is in the state where $Q_0Q_1Q_2Q_3 = 1001$, and $D_0D_1D_2D_3 = 1010$ is loaded in. When the $SHIFT/\overline{LOAD}$ input is taken HIGH, the data shown in Figure 8–63 are shifted out. Is this operation correct? If not, what is the most likely problem?

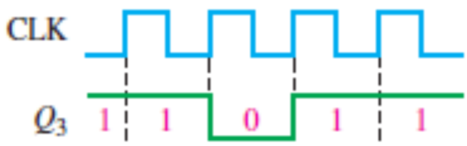


FIGURE 8–63

Since the LSB flip-flop works during serial shift, the problem is most likely in gate G3. An open D_3 input at G3 will cause the observed waveform. See Figure 8-23.

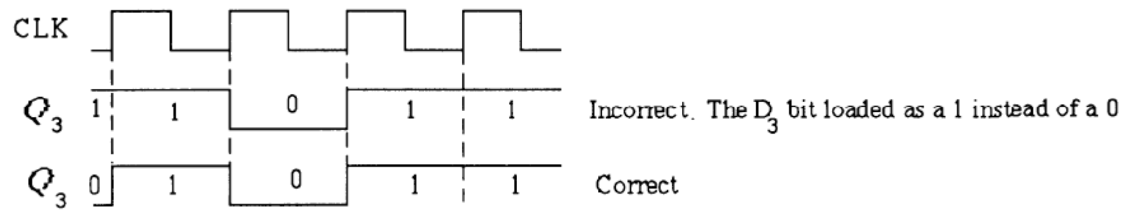


FIGURE 8-23